Candidate's name/ Index number ……

THE UNITED REPUBLIC OF TANZANIA
THE PRESIDENT'S OFFICE
REGIONAL ADMINISTRATION AND LOCAL GOVERNMENT
LONGIDO DISTRICT
ANNUAL JOINT EXAMINATION

FORM ONE
032 CHEMISTRY

Time: 2½ Hours **November 2024**

Instructions:

1. This paper consists of sections A and B with a total of ten (10) questions.
2. Answer ALL questions in the spaces provided.
3. All writing must be in blue or black ink, except drawing, which must be in pencil.
4. Cellular phones and any unauthorized materials are **not** allowed in the examination room.
5. Write your **Examination Number** at the top right corner of every page.

QUESTION #	MARKS	INITIALS OF EXAMINER
1		
2		
3		
4		
5		
6		
7		
8		
9		
10		
Total		

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SECTION A (15 Marks)

Answer all questions

1. Choose the most correct answer among the given alternatives and write down the answer in the space given.

(i) One of the following is NOT a purpose of first aid
- A. To relive pain
- B. To give hope
- C. To prevent permanent disability
- D. To cure diseases like malaria

(ii) When solid iodine is heated, the following occurs:
- A. It evaporates into gaseous state
- B. It melts into a liquid iodine
- C. It decomposes into a liquid substance and gas
- D. It remains solid because of its high melting point

(iii) Bunsen burner is preferred over kerosene burner for heating in the Chemistry laboratory because
- A. It has low heat which is important in preventing test tubes from breaking
- B. It can produce luminous flame
- C. It is cheaper than the kerosene burner, hence affordable
- D. It is convenient to handle and produces hot flame

(iv) The part of the Bunsen burner that directs gas to the barrel is called:
- A. Air hole
- B. Collar
- C. Jet
- D. Chimney

(v) A student at Seypalm secondary school trying to heat chalk so that it can decompose into calcium oxide and carbon dioxide. Which of the following equipment should he use?
- A. Motor and pestle
- B. Crucible
- C. Petri dish
- D. Gas jar

(vi) Which of the following statements is correct?
- A. Gases are less compressible than liquids
- B. Gases are incompressible
- C. Solids are more compressible than liquids
- D. Gases are more compressible than liquid

(vii) Which of the following describes better the process of melting>
- A. Molecules lose energy and becomes more mobile
- B. Molecules lose energy and becomes less mobile

C. Molecules gain energy and become more mobile
D. Molecules gain energy and become less mobile

(viii) The step that follows after hypothesis formulation in the scientific procedure is
A. Data analysis
B. Experimentation
C. Conclusion
D. Problem question

(ix) Which of the following set of symbols represent potassium, sodium, and carbon respectively?
A. K, Na and C
B. P, S and Ca
C. K, S and Ca
D. P, Zn and C

(x) You want to measure exactly 20 mililiters of a solution. Which of the following measuring apparatus would be more appropriate to use?
A. Measuring cylinder
B. Pipette
C. Beaker
D. Burette

ANSWERS

List A	(i)	(ii)	(iii)	(iv)	(v)	(vi)	(vii)	(viii)	(ix)	(x)
List B										

2. Match the mixtures in list A with the correct separation method in list B.

LIST A	**LIST B**
(i) Prevents infection (ii) Disinfects wounds (iii) Soothes broken skin (iv) Splinter removal and securing bandages (v) Reducing muscle pain	A. Petroleum jelly B. Bandaged C. Safety pin D. Liniment E. Antibiotic F. Distilled water G. Soap H. Spirit I. Pair of scissors

ANSWERS

List A	(i)	(ii)	(iii)	(iv)	(v)
List B					

SECTION B (70 Marks)

3. (a) When a teacher was talking to students at parade ground, suddenly one student fell down; teachers and students rushed to help the student.

(i) Mention and define the process they were going to deliver to the student

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(ii) Mention and define the collection of accessories the teachers and students who rushed to the student who fell down would need.

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(b) State four importance of the process defined in (a)(i)

(i) ……

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(ii) ……

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(iii)……

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(iv)……

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4. Flames are used for different purposes. Briefly explain at least four uses of the flames.

(i) ……

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(ii) ……

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(iii)……

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(iv)……

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5. (a) What is an element?

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(b) (i) Each element has been assigned a shorthand to represent it. With one accompanying example, state three rules used when doing that.

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- ……

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- ……
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(ii) Write symbols for each of the following elements:

Name	Symbol
Lithium	
Sodium	
Potassium	
Magnesium	

6. (a) Explain the components that would make a fire continue or start to burn when they are present:

(i) ……
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(ii) ……
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(iii)……
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(b) A Chemistry teacher said, burning and rusting are basically similar although they can be easily distinguished; explain what the teacher meant.

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7. Give differences between seawater and pure water in terms of the properties of their components or constituents, their composition characteristics, nature of their formation, and properties of the substances compared to the properties of their constituents.

(i)	
(ii)	
(iii)	
(iv)	

8. Giving five points, explain why without the ability of matter to change from one state to another, life on Earth would have been almost impossible.
   (i) ……
   ……
   (ii) ……
   ……
   (iii)……
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   (iv)……
   ……
   (v) ……
   ……

9. (a) Whenever visiting Chemistry laboratory, Seypalm noticed the doors always opened outwards there was a fume chamber inside. Briefly explain the reasons for the doors working that way and the presence of the observed structure.
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   (b) Explain four differences between solutions and suspensions.

   (i) ……
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   (ii) ……
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(iii)

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(iv)

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SECTION C (15 MARKS)

10. The knowledge of chemistry, like other science, is obtained by using the scientific procedure. Explain this process to your fellow students by telling them what the scientific procedure actually is and by giving at least, six ways the scientific procedure is important.

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